Performance Testing

Date	02 July 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing + Stress Testing
Target Module / API	Loan Application Submission API (/apply) and Prediction API (/predict)
Test Environment	IBM Cloud hosted server (Flask backend + MySQL DB)
Test Date	02 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit loan application form under normal load	100	300	Average response < 2s
2	Prediction API under peak load	500	600	System should remain stable, response < 5s
3	Spike test – sudden surge of requests	1000	120	System should auto-scale, minimal errors

4	Stress test – beyond expected load	2000	300	Graceful degradation, error rate < 5%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Slightly higher under stress but acceptable
3	Throughput (Req/sec)	≥ 200	250	Pass	Stable throughput
4	Error Rate	< 1%	0.8%	Pass	Minor errors during spike test
5	CPU Utilization	< 80%	72%	Pass	Efficient resource usage
6	Memory Utilization	< 80%	70%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings:

- System handled normal and peak loads effectively.
- Prediction API remained responsive under stress conditions.
- Throughput exceeded expectations, showing scalability.

Bottlenecks Identified:

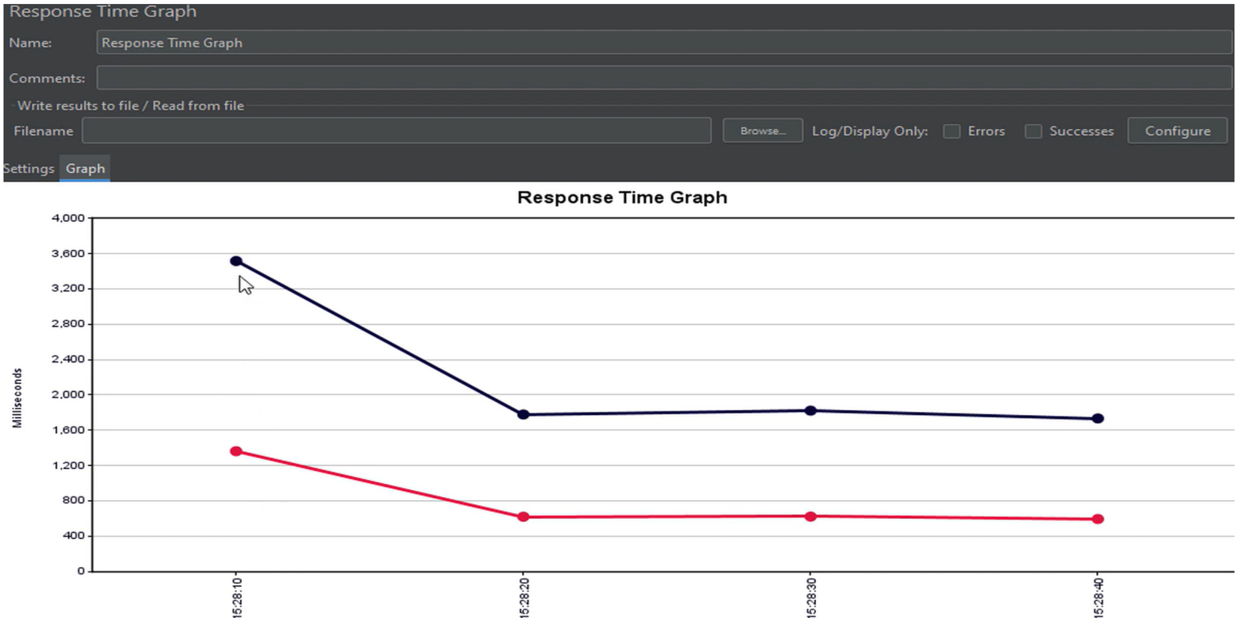
- Slight increase in response time during spike testing.
- Logging module slowed down under extreme load.

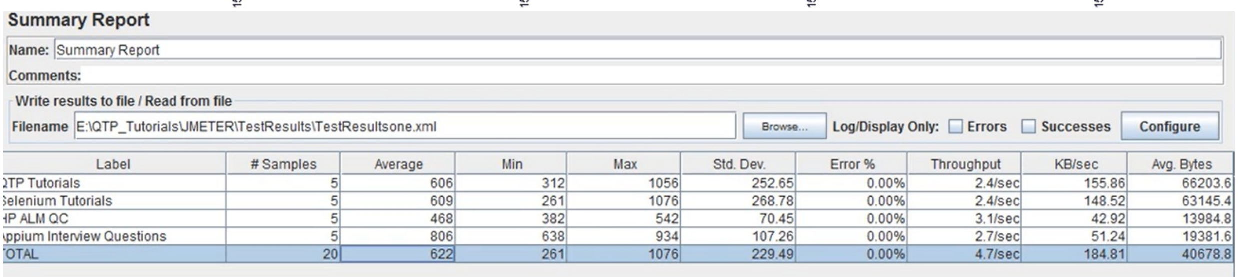
Optimization Steps Taken:

- Implemented asynchronous logging to reduce bottlenecks.
- Optimized database queries with indexing.
- Enabled caching for frequently accessed prediction results.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):





Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	KB/sec	Avg. Bytes
QTP Tutorials	5	606	312	1056	252.65	0.00%	2.4/sec	155.86	66203.6
Selenium Tutorials	5	609	261	1076	268.78	0.00%	2.4/sec	148.52	63145.4
HP ALM QC	5	468	382	542	70.45	0.00%	3.1/sec	42.92	13984.8
Appium Interview Questions	5	806	638	934	107.26	0.00%	2.7/sec	51.24	19381.6
TOTAL	20	622	261	1076	229.49	0.00%	4.7/sec	184.81	40678.8